

INTELLIGENT FRAUD PREVENTION & INVESTIGATION REPORT

HIGH RISK

Recommendation: ACTION REQUIRED: SUSPEND TRANSACTION & CONTACT CARDHOLDER

RISK SCORE

80.00 / 100

MODEL DECISION

POTENTIAL FRAUD

Transaction Overview & Account Baseline

Current Transaction	Evaluated Values	Customer Profile Baseline	Historical Norms
Transaction ID	TXN0000211066	Historical Total Txns	0
Customer ID	CUST00199334	Average Spend	\$105.84
Timestamp	2024-12-30 23:56:00	Spend Std Deviation	\$85.73
Amount	\$15.62 USD	Median Spend	\$94.95
Location	Melbourne, India	Home City	Manchester
Category	Education	Home Country	Canada
Payment Method	Bank Transfer	Primary Category	Clothing
Device Type	ATM	Primary Payment	Bank Transfer
Distance from Home	7.3 km	Primary Device	Mobile
Time Since Prior Txn	14.93 hrs	Avg Transaction Hour	17.6:00 hrs
Failed PIN Attempts	0	Txn Velocity	0.01 / day
Recent PIN Change	No	Customer Age	65 yrs
Cross-Border / Intl	No (Domestic)	Credit Score	528
Night Window	Yes (Off-hours)	Prior Confirmed Frauds	0

AI Model Risk Breakdown & Explainability

Autoencoder Anomaly Score		0.2980 (0.00 – 1.00)		Reconstruction MSE Error		0.353351 (Baseline Threshold: 0.5929)	
Risk Component	Score	Weight	Contrib (pts)	AI Explanation & Context			
Geographic Risk	1.00	15%	+15.00	■ IMPOSSIBLE TRAVEL DETECTED: 16970 km from Manchester to Melbourne in 895.8 minutes (implied speed: 1137 km/h). Physical travel is impossible at this speed.			
Autoencoder	0.30	35%	+10.43	Transaction pattern appears relatively normal (autoencoder score: 0.30).			
Time Anomaly	0.70	10%	+7.00	Transaction at 23:00 — unusual off-hours activity (night transaction flag set).			
Velocity Risk	0.19	10%	+1.87	Transaction 14.9 h after last — within normal velocity range.			
Behavioral Deviation	0.18	8%	+1.42	Behavioural deviations detected: unusual city 'Melbourne', unusual category 'Education', unusual device 'ATM'. Customer normally uses Bank Transfer in Manchester at avg \$106.			
Credit Factor	0.49	1%	+0.49	Poor credit score (528) — elevated credit risk.			
Amount Deviation	0.01	15%	+0.16	Transaction amount \$15.62 is 1.1σ below customer's average \$105.84 (σ=\$85.73).			
International Flag	0.00	3%	+0.00	Domestic transaction — no international risk.			
Failed Attempts	0.00	2%	+0.00	No prior failed attempts on this transaction.			

Pin Change	0.00	1%	+0.00	No recent PIN change.
------------	------	----	-------	-----------------------

Audit & Compliance Sign-Off

Investigator Signature:

Verified Date: 2026-10-07

This report is automatically synthesized by the SENTINEL Deep Autoencoder engine with 10-factor weighted anomaly scoring. All data points conform to regulatory audit guidelines.